CONCEPT 01 · INTER-AGENT EXCHANGE

Four exchange media — only one on by default

At each layer, the “information exchanged” changes nature — and maturity decreases as the agent becomes more “agentic”.

Layer 1 · latent vector

tensor rows read each other
via N×N attention (a matmul)

SERVED by default

Layer 2 · text + receipt

Claude / GPT / Gemini exchange answers,
sealed under a re-verifiable receipt

real · OFF by default

4 independent binaries — no shared data path at runtime

Layer 3 · state vector (gRPC)

swarm-operators Φ coupled through a
meta-consensus circulating as a “pull”

MOCK κ·x by default

Layer 4 · JSON cells + bus

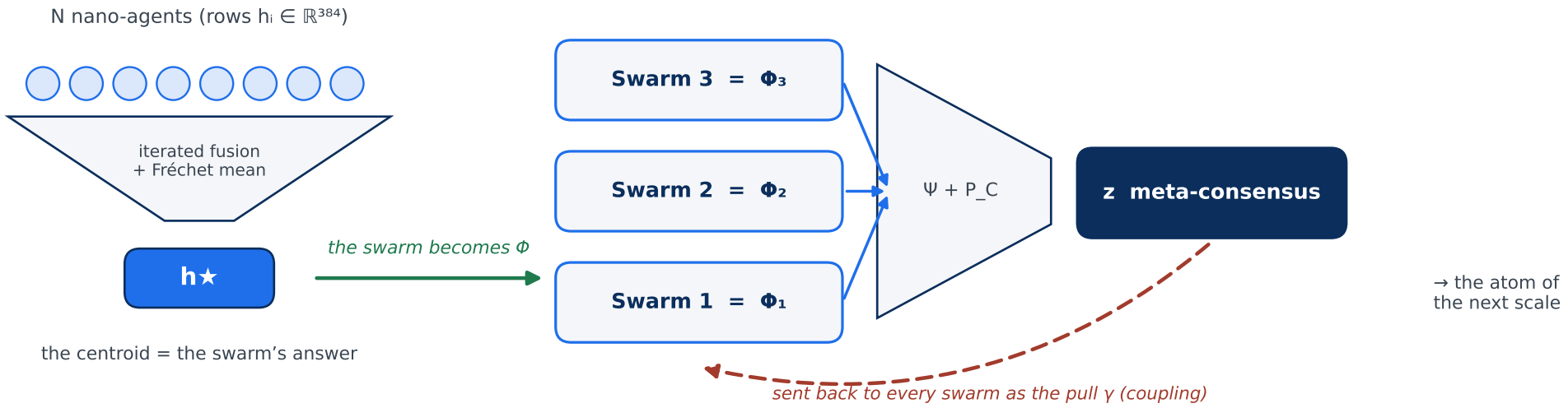
DAG nodes write to a versioned
blackboard + EventBus

CANNED without a key

Conceptual diagram — the maturity shown is the REAL default state (rule: never over-sell); details: EFE reports, 2026-07-14.

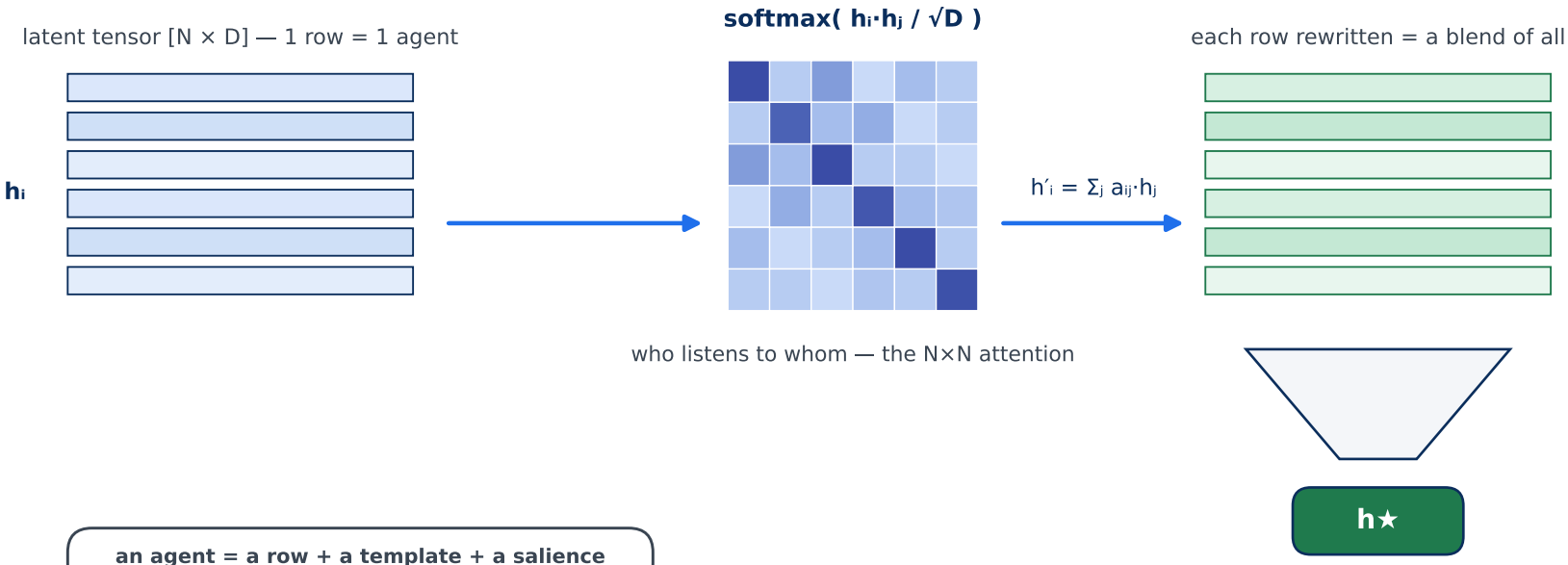
The vertical fold — a population becomes a vector

The same gesture at every scale: N voices → 1 vector, which becomes the atom of the scale above.



An agent’s voice is a vector

No messages, no negotiation: each agent reads all the others through attention, then rewrites its own row.

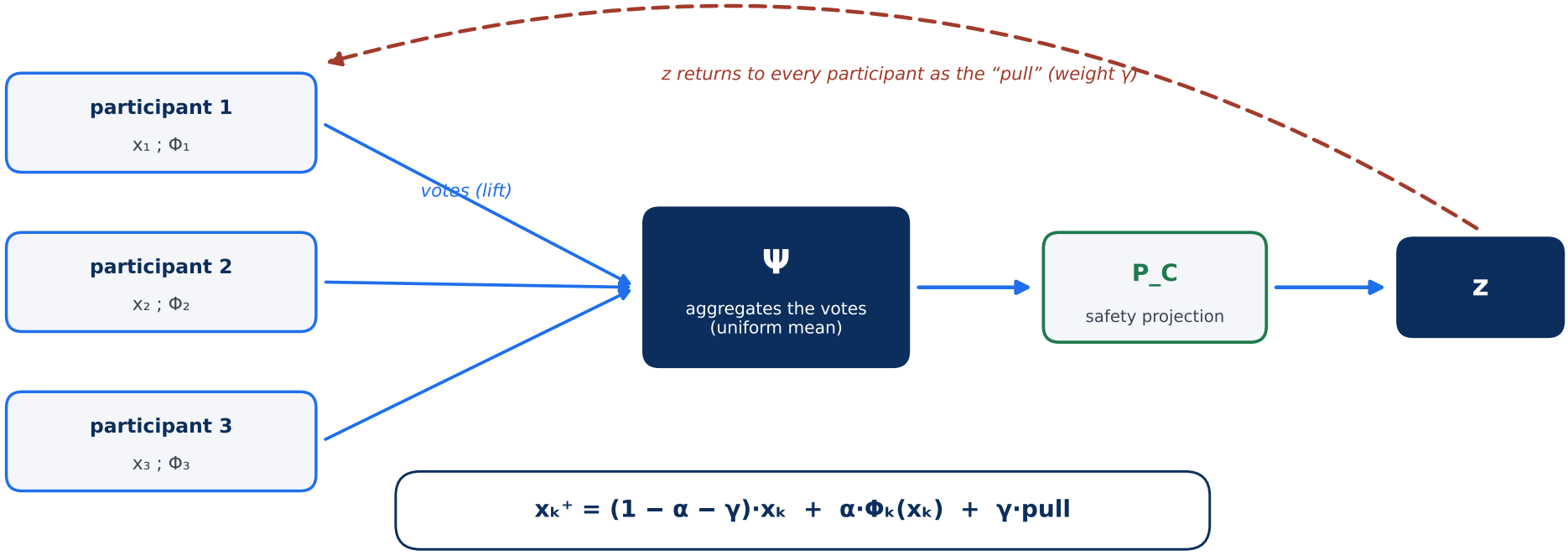


an agent = a row + a template + a salience

template priors “Not meaningful” — the signal = the injected embedding

The coupled fixed point (Krasnoselskii–Mann)

The meta-swarm exchanges exactly one thing: a state vector — fused into a consensus, sent back as a pull, until the fixed point.



$$x_k^+ = (1 - \alpha - \gamma)\cdot x_k \;+\; \alpha\cdot\Phi_k(x_k) \;+\; \gamma\cdot \text{pull}$$

contracts if H1 holds ($L_\Phi \le 1$) — hence the probe of concept 05

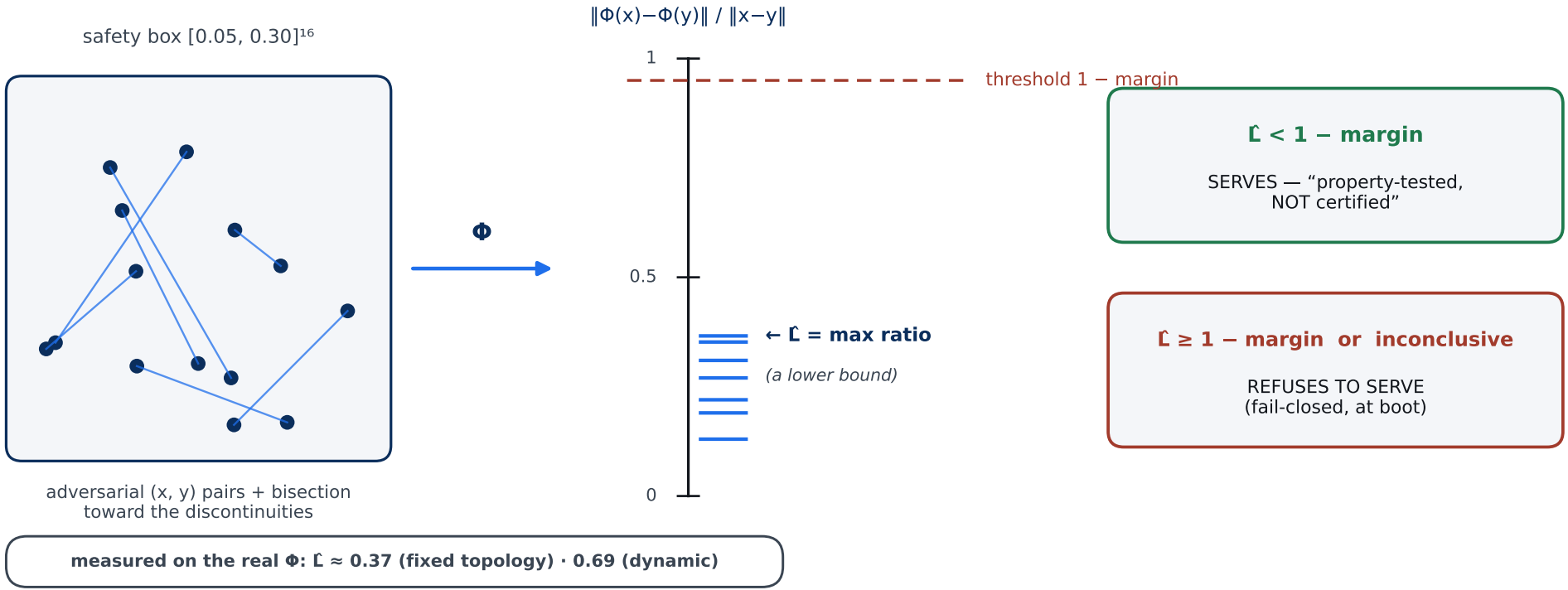
by default: Φ_k = mock $\kappa\cdot x$ — the real swarm-as-Φ is opt-in (M1)

Ψ = a mean: no robustness

Concept: the Layer-3 coupled iteration (our M1 work). Safety is projected twice per step (participant output + meta-consensus).

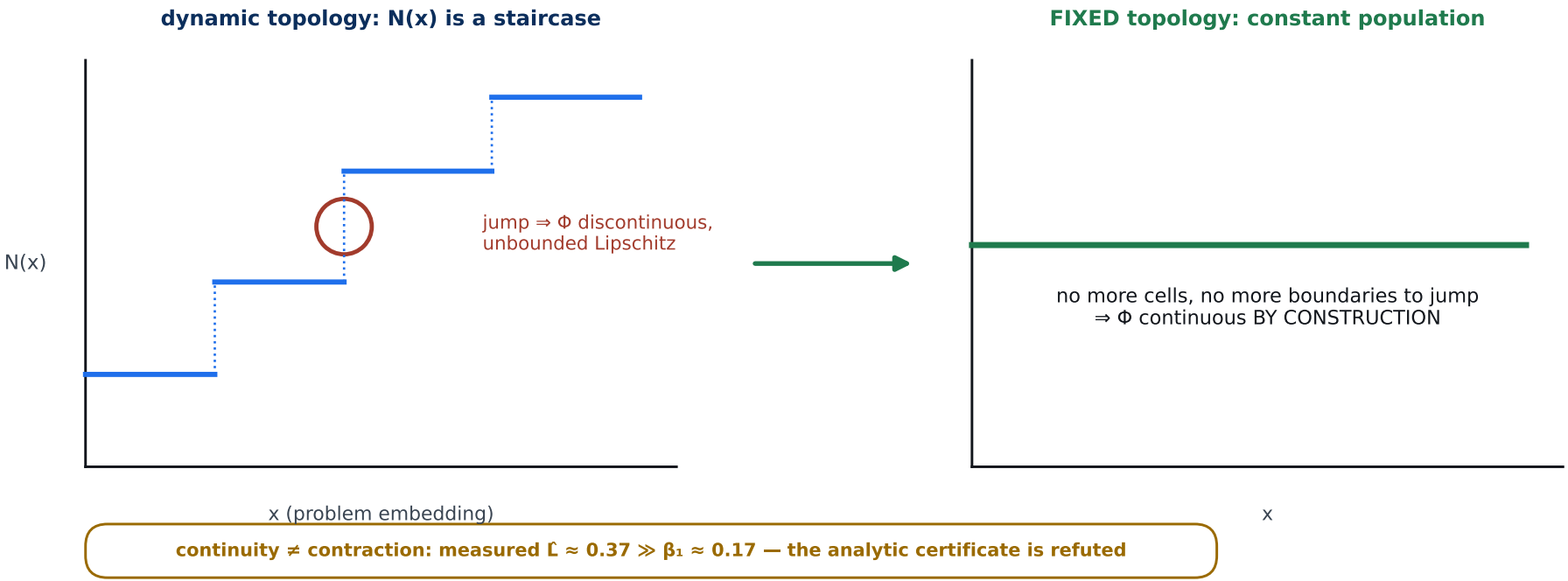
The H1 probe — falsify, never certify

Before coupling a swarm to the meta-swarm, we attack its operator Φ: the probe measures a LOWER bound L̂ of its sensitivity.



Removing the discontinuity — the fixed topology

The coupling's real enemy is not Φ's slope: it is its JUMPS — genesis routing turns x into an INTEGER number of agents.



Fail-closed — every gate shuts by default

The principle running through all our M1 work: between “staying silent” and “possibly lying”, the system always chooses to refuse.

manifest →

H1 canary at boot

the probe attacks Φ BEFORE the port is bound

Ľ too high / inconclusive ⇒ does not serve

→

Engine identity

config_hash on the wire; kappa/seed → Option

old binary ⇒ parse failure — never a silent mock

→

Convergence required

require_convergence: not converged ⇒ ABORTED

never a non-converged “consensus” returned OK

→

Lineage on reload

an id’s engine identity is frozen for the process lifetime

remove-then-re-add ⇒ rejected

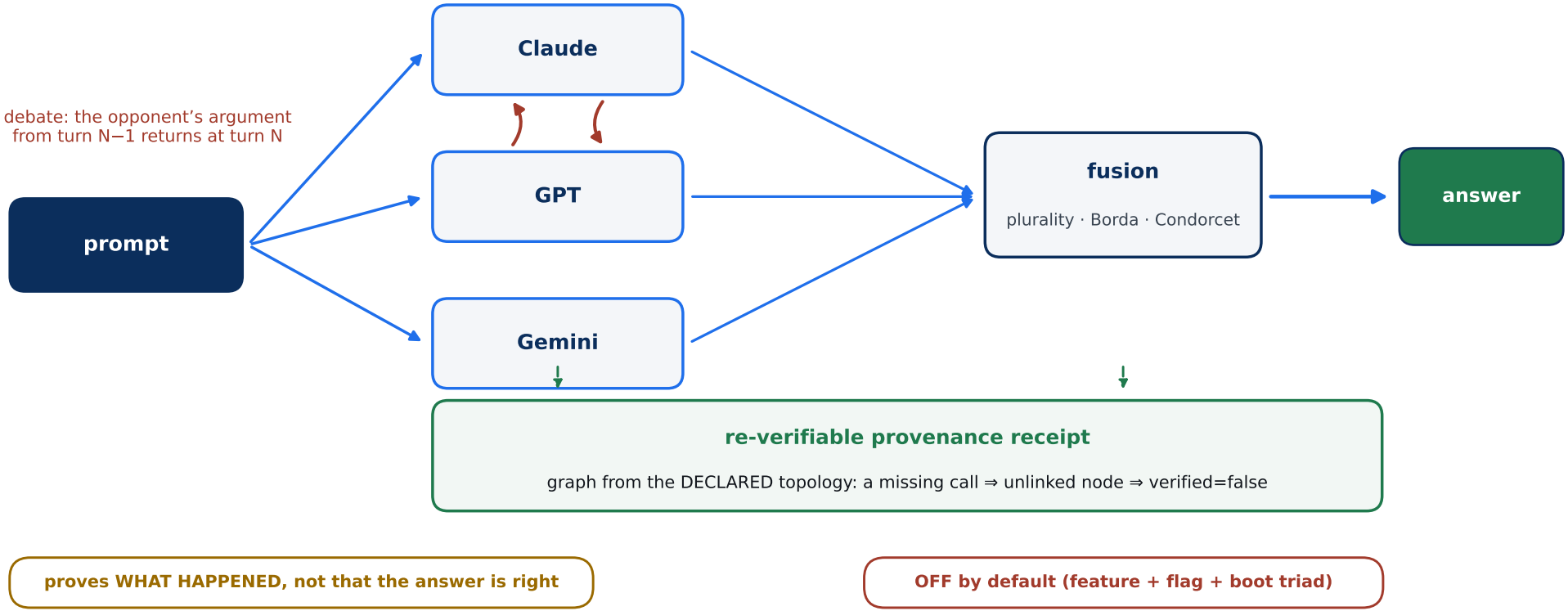
→ **serve**

the default is never “pretending” — the default is REFUSING

Concept: the meta-swarm’s four fail-closed gates (M1.c/M1.d/665ead0). The 4th closes the remove-then-re-add bypass found in adversarial review.

The frontier panel — text, under a re-verifiable receipt

The only layer where agents exchange LANGUAGE — and only the “debate” mission circulates real arguments.



Concept: Layer 2. Honesty: Borda = a heuristic (self-ranking + Jaccard); vote/relay/specialist/cascade do not exchange arguments.